Scenario&Architecture&Requirements of the IoA

Chong Zheng (zhengchong6@huawei.com)
Nan Geng (gengnan@huawei.com)

IETF 125 – AI Agent Policy-aware Network@Shenzhen, Mar 18 2026

Content

Digital Banking Scenario

- focusing on the bank union member interconnection case
- based on an on-going joint innovation project with UnionPay

Brief Introduction of IoA (Internet of agents) Layered architecture

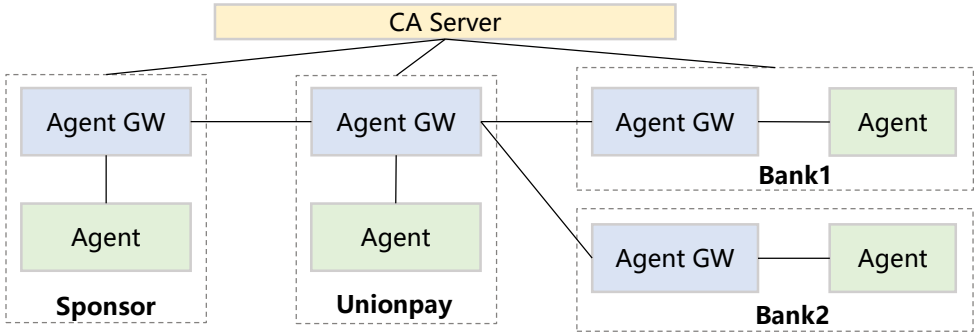
- Application Layer
- Agent Management Layer
- Agent Interconnection Layer
- Underlay Network Layer

Fine granular network service requirements for agent communication

- Fine-grained **SLA** network service requirements for agent communication
- Fine-grained **security** network service requirements for agent communication
- Fine-grained **advanced** network service requirements for agent communication

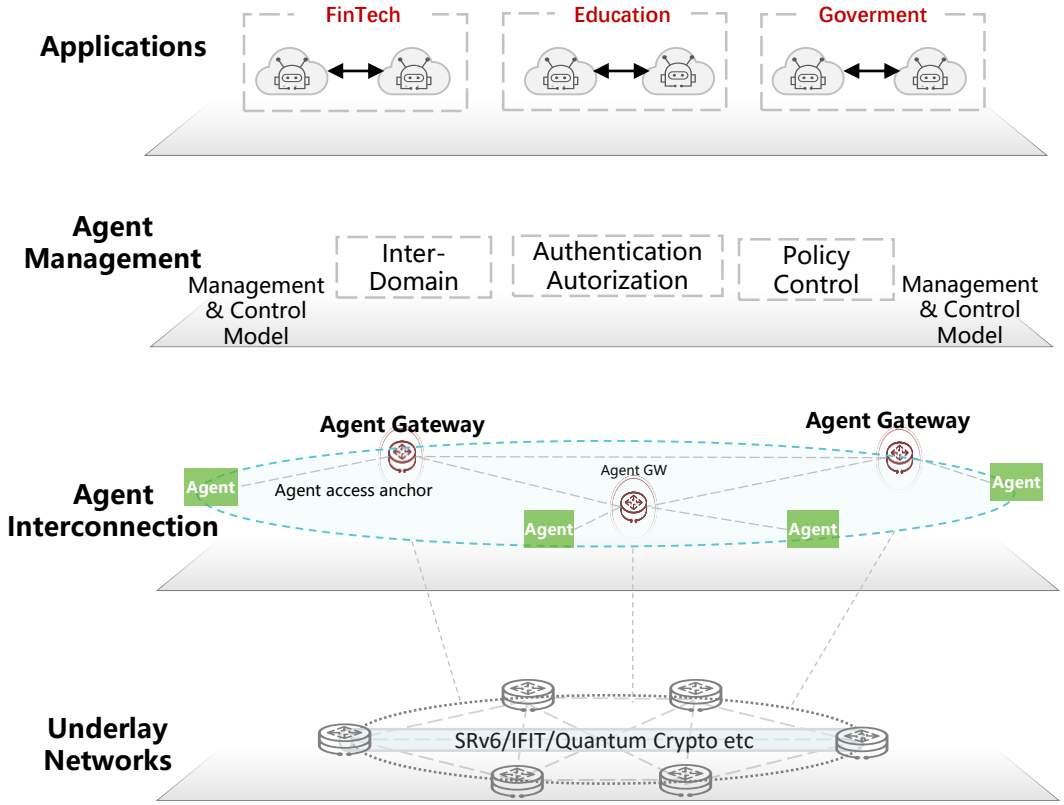
Digital banking Scenario

We have implemented intelligent open banking business scenarios based on Agent and Agent Gateway, and simplified the interconnection between application parties.



- **Sponsor:** Provide users with the ability to query bank transaction records.
- **Unionpay:** Shield the differences between banks and simplify the interaction between the Sponsor and different banks.
- **Bank:** Return the corresponding user transaction information based on the request.

Layered architecture abstraction of IoA (Internet of Agents)



The agent application layer consists of many specific agent applications.

The agent management layer implements network management and control for agent communication , Including

- Policy control
- Authentication & Autorization
- etc.

The networking between agent gateways is a new plane built upon the underly network infrastructure.

Stronger network assurance capabilities are required for communication between AI Agents, which can be achieved through the underlay network.

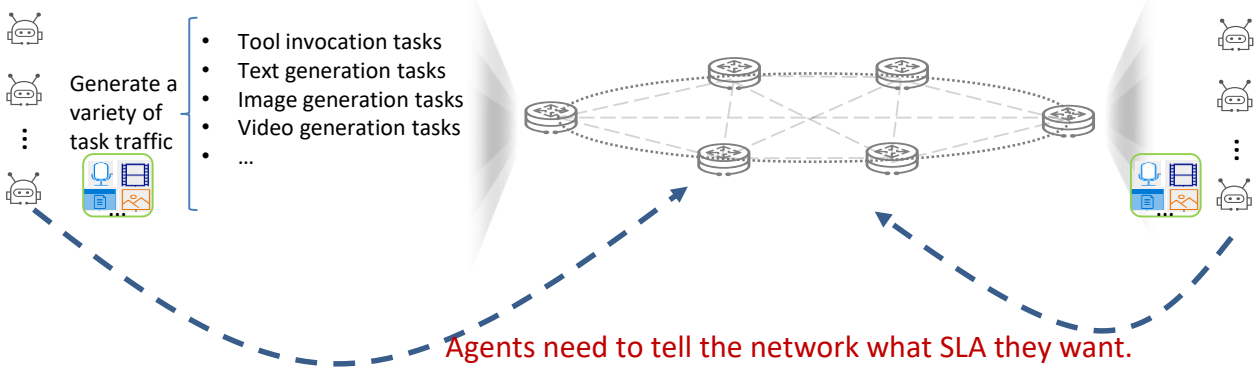
- SRv6
- IFIT
- Quantum Crypto
- etc.

Underlay synergy: Agents communication needs fine-grained **SLA** services from network

1. Interaction of Agents needs Differentiated SLA Guarantee from network

- **Tool invocation** tasks (real-time interaction): **High** sensitivity to latency and jitter, **Low** bandwidth
- **Text generation** tasks: **Medium** sensitivity to latency, **Low** bandwidth.
- **Image generation** tasks: **Low** sensitivity to latency, **Medium** bandwidth demands.
- **Video generation** tasks: **Low** sensitivity to latency and jitter, **Very High** bandwidth demands.

➔ **Agents need to tell the network what SLA they want.**

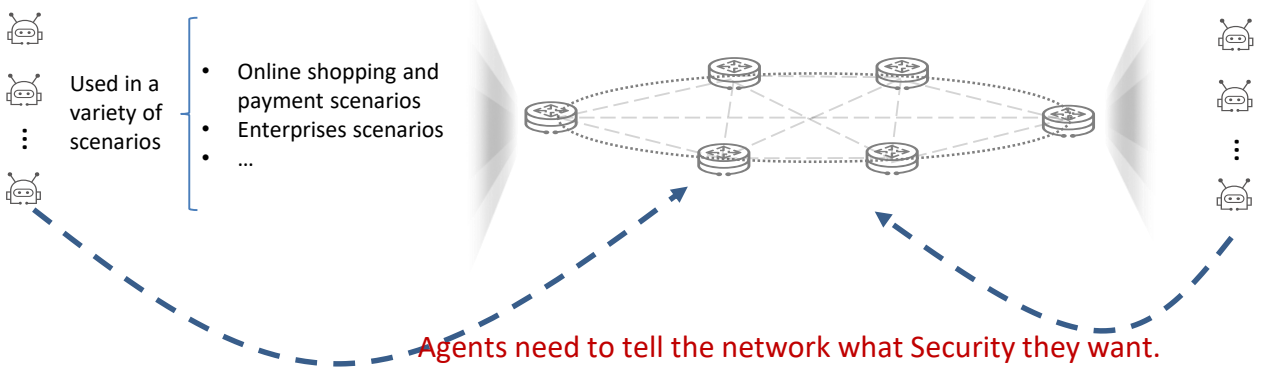


Underlay synergy: Agents communication needs fine-grained **security** services from network

2. Agents need Secure Transmission from network

- **Online shopping and payment scenarios:** personal account information and financial data require higher level of security for transmission.
- **Enterprises scenarios:** some agents that can access sensitive data are not allowed to collaborate with external agents to prevent data leakage.

➔ **Agents need to tell the network what Security they want.**

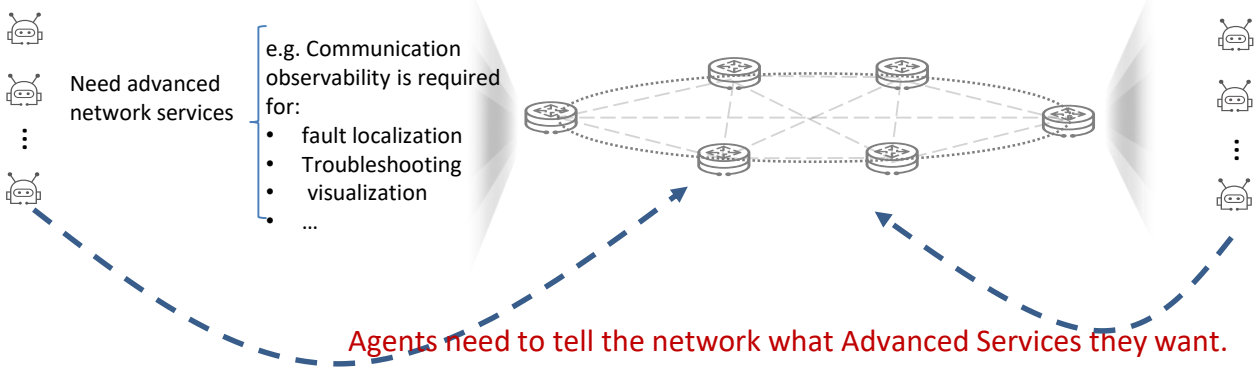


Underlay synergy: Agents communication needs fine-grained **advanced** services from network

3. Agents need Measurement and Visualization from network

- The quality of communication may need to be measured for purposes such as fault localization, troubleshooting, and visualization.

➔ **Agents need to tell the network what Advanced Services they want.**



Thank You!

Further reading:

- https://datatracker.ietf.org/doc/draft-men-rtgwg-agent-networking-in-digibank/
- https://datatracker.ietf.org/doc/draft-zlgsgl-rtgwg-agents-networking-framework/
- https://datatracker.ietf.org/doc/draft-mao-rtgwg-apn-framework-for-ioa/